

**REAL-TIME WEARABLE SMART SYSTEM FOR
DETECTING PATELLAR DISORIENTATION AND
ENHANCING PATELLAR STABILITY**

A MINI-PROJECT REPORT

Submitted by

HARINISRI RAMESH - 3122233003018

MAANASVINI ANAND - 3122233003028

KAYALVIZHI M - 3122233003023

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Sri Sivasubramaniya Nadar College of Engineering

(An Autonomous Institution, Affiliated to Anna University)

Rajiv Gandhi Salai (OMR), Kalavakkam – 603 110

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BONAFIDE CERTIFICATE

Certified that this project titled “**Real-Time Wearable Smart System For Detecting Patellar Disorientation And Enhancing Patellar Stability**” is the bonafide work of Harinisri Ramesh, Maanasvini Anand and Kayalvizhi M, who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

Dr. A. Kavitha

HEAD OF THE DEPARTMENT

Head of the department

Department of Biomedical

Engineering

SSN College of Engineering

Kalavakkam - 603110.

Dr. S. Saranya

SUPERVISOR

Assistant Professor

Department of Biomedical

Engineering

SSN College of Engineering

Kalavakkam - 603110.

Submitted for project viva-voce examination held on

EXTERNAL EXAMINER

INTERNAL EXAMINER

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HARINISRI RAMESH - 3122233003018

MAANASVINI ANAND - 3122233003028

KAYALVIZHI M - 3122233003023

ABSTRACT

Patellar instability, commonly arising from muscle imbalance or prolonged inactivity, can lead to conditions like Patellofemoral Pain Syndrome (PFPS) if not addressed in its early stages. Conventional braces can help provide patellar alignment, but they are rigid and do not offer feedback as they are passive systems. This project aims to bridge this gap by developing an attachable smart sensor module that can upgrade conventional knee braces with real-time biomechanical monitoring capabilities

At its core is a lightweight, custom 3-D printed knee brace which is anatomically contoured. It includes three essential components: an ESP32 WROOM micro-controller, an MPU6050 motion sensor, and a thin strip of piezoelectric PVDF film that detects the patellar vibrations. The ESP32, programmed in the Arduino IDE, sends angular data from the motion sensor and vibration patterns from the film sensor directly to a custom web app. The application allows real-time monitoring by displaying knee flexion and extension angles, along with patellar vibrations and warns them whenever an unusual patellar vibration is observed. This feedback mechanism allows wearers to adjust the brace accordingly.

This system provides dynamic biomechanical feedback, improving the accuracy of patellar tracking, rehabilitation assessment, and early detection of instability. By merging additive manufacturing, embedded sensing, and IoT-based data analytics, the project presents an accessible and adaptive solution to enhance rehabilitation outcomes in patients with patellar disorientation.

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CHAPTER 1

INTRODUCTION

1.1 PROBLEM STATEMENT AND NEED FOR SOLUTION

The knee joint is one of the most complex and load-bearing joints in the human body. It enables locomotion, supports body weight during movement, and ensures overall postural stability. The proper alignment and tracking of the patella are essential for maintaining efficient biomechanics of the knee. However, sedentary lifestyle and muscle weaknesses lead to abnormalities such as patellar disorientation or lateral displacement. These conditions often lead to joint instability, discomfort, pain, and restricted mobility. Over time, these conditions can contribute to chronic knee dysfunction, early-onset osteoarthritis, or irreversible cartilage degeneration. Hence, maintaining patellar alignment is vital for preventing both acute and degenerative knee pathologies.

Conventional knee braces provide structural support and limit undesired movement, thereby reducing the risk of further injury. However, these conventional designs are passive in nature and offer only mechanical stability without supplying any form of real-time biomechanical feedback. This static nature makes it difficult to detect subtle changes in patellar motion, variations in joint pressure, or abnormal angular deviations that occur during rehabilitation or daily activities. Without quantitative feedback, clinicians and patients are unable to evaluate the effectiveness of therapy or monitor early signs of malalignment, resulting in delayed diagnosis and inefficient recovery.

In this project, we present a **Real-Time Wearable Smart System for Detecting Patellar Disorientation and Enhancing Patellar Stability**, which combines the principles of biomechanics, embedded electronics, and additive manufacturing. The proposed knee brace is custom-designed using 3D printing technology based on the curvature of an existing anatomical knee model, allowing for a personalized and ergonomic fit. The brace integrates an **ESP32 WROOM microcontroller** as the core processing unit, an **MPU6050 inertial measurement unit (IMU)** to monitor joint angles and acceleration, and a **PVDF piezoelectric film sensor** to capture minute patellar vibrations. The signals obtained from these sensors are processed by the ESP32 and wirelessly transmitted via Wi-Fi to a **web application**.

The web interface serves as an intelligent monitoring platform, capable of visualizing real-time vibration and angular data. It incorporates calibration features that establish normal operating thresholds and automatically alert users when readings deviate from expected limits. These warnings can indicate potential patellar misalignment, excessive joint stress, or abnormal vibration patterns that precede injury. Through this approach, both clinicians and users can monitor rehabilitation progress remotely, adjust physical therapy routines, and ensure safer and more effective recovery.

Furthermore, the use of 3D printing provides several engineering advantages, such as precise sensor embedding, weight reduction, material flexibility, and customization for different patient anatomies. By combining mechanical adaptability with electronic sensing, this project aims to bridge the gap between traditional passive orthoses and modern, data-driven wearable devices. The ability to measure, transmit, and interpret biomechanical parameters dynamically enhances clinical assessment and empowers users with continuous insights into their knee health.

1.2 LITERATURE SURVEY

1. An Intelligent Non-Invasive Blood Pressure Monitoring System Based on Modified Piezoelectric Polymer Authors: S. Li et al

METHODOLOGY: Developed a flexible PVDF-based sensor system for wearable blood pressure monitoring on the wrist.

FINDINGS: High sensitivity and flexibility suitable for real-time pulse wave detection; required signal conditioning to reduce noise.

LIMITATIONS: Susceptible to electromagnetic interference affecting signal clarity.

2. Design and Structural Analysis of a Low-Cost 3D-Printed Knee Orthosis with Embedded Sensors F. Popișter et al.

METHODOLOGY: Developed a 3D-printed knee orthosis tailored to patient anatomy, embedded with IMUs and EMG sensors for real-time movement and muscle activation feedback

FINDINGS: Demonstrated effective gait correction and postural improvement with real-time sensor feedback.

LIMITATIONS: Complexity in sensor calibration and integration posed challenges.

3. A Wearable, Multimodal Sensing System to Monitor Knee Joint Health RSC Advances, 2024

METHODOLOGY: Developed a knee brace with contact microphones, EBI, and IMUs to collect acoustic, swelling, and inertial data for joint health assessment.

FINDINGS: Enabled multimodal sensing and joint health scoring for remote monitoring.

LIMITATIONS: Wearability and integration require improvement; experimental validation phase

4. Development of a multi-modal sensor network to detect and analyze acoustic emissions from knee joints, Sensors and Actuators A: Physical, 2022

METHODOLOGY: Created and characterized a sensor system including piezoelectric sensors for detecting acoustic signatures related to knee joint health

FINDINGS: The system captured subtle joint acoustic emissions useful for early detection of mechanical changes.

LIMITATIONS: Primarily experimental with limited clinical validation

5. Development of a multi-modal sensor network to detect and analyze acoustic emissions from knee joints

METHODOLOGY:

A multi-sensor system using piezoelectric acoustic sensors was developed to capture vibration and sound signals from the knee during flexion–extension. Sensor placement, signal acquisition, and filtering methods were optimized to isolate joint-generated acoustic emissions.

FINDINGS:

The system detected subtle acoustic signatures linked to early mechanical changes in the knee. Differences in signal patterns between smooth and irregular joint motion demonstrated the potential of acoustic sensing for early detection of cartilage or stability issues.

LIMITATIONS:

The study was experimental with a small sample size and limited clinical validation. Testing occurred in controlled conditions, and the wearable integration of the system requires further development.

CHAPTER 2

METHODOLOGY

2.1 COMPONENTS

2.1.1 COMPONENTS OVERVIEW

The MPU6050 inertial measurement unit (IMU) is used to capture raw angular and accelerometric data corresponding to the knee joint's motion during flexion and extension. Which are read as analog motion data and then processed digitally by the ESP32-WROOM microcontroller. In parallel, a PVDF-based piezoelectric film sensor is embedded near the patellar region to detect micro-vibrations produced due to patellar tracking irregularities. The PVDF generates low-amplitude electrical signals proportional to surface vibration intensity, which are similarly acquired and converted for digital interpretation within the ESP32. Both the IMU angle data and PVDF vibration data are wirelessly transmitted via the ESP32's inbuilt Wi-Fi module to a custom web-based application for further visualization and monitoring. All the hardware components are then housed in a custom 3D printed knee brace which allows for efficient and ergonomic data acquisition.

SL.NO	COMPONENT	SPECIFICATION	QUANTITY
1	Microcontroller	ESP32 WROOM	1
2	IMU	MPU6050	1
3	Piezoelectric sensor	PVDF Film sensor	1
4	Knee Brace	Custom 3D Printed brace	1
5	Passive Components	4.7 k Ω Resistor	2
6	Jumper Wires		As required

2.1.2 COMPONENTS USED

ESP32 WROOM Microcontroller

Acts as the core processing and communication unit, featuring dual-core performance, integrated Wi-Fi, and Bluetooth for wireless data transmission.

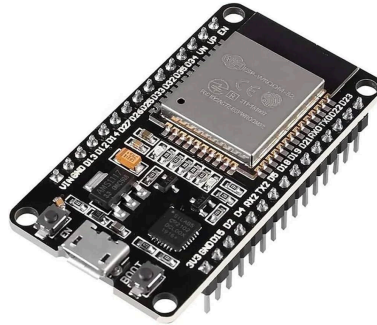


Fig 2.1 ESP32 WROOM 32

PVDF film sensor

A PVDF (Polyvinylidene Fluoride) film sensor is a flexible piezoelectric material that generates voltage when subjected to mechanical stress or vibration. In biomedical applications, it is commonly used to detect subtle physiological vibrations, such as patellar movement or impact forces. The sensor converts these mechanical vibrations into low-amplitude electrical signals, which are then conditioned and fed into a microcontroller for analysis. Since PVDF responds to dynamic vibration changes, it is ideal for real-time biomechanical feedback applications.



Fig 2.2 PVDF filmSensor

IMU (MPU6050)

The MPU6050 is a MEMS-based inertial measurement unit (IMU) that combines a 3-axis accelerometer and a 3-axis gyroscope in a single compact module. It measures linear acceleration and angular velocity, allowing precise tracking of motion and joint orientation. The sensor communicates digitally with the microcontroller using the I2C protocol and provides real-time angle and movement data.

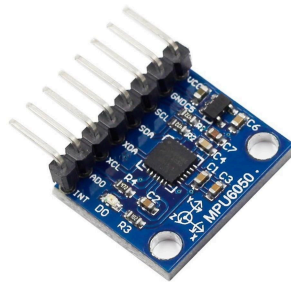


Fig 2.3 MPU6050

CUSTOM 3D PRINTED KNEE BRACE

The custom knee brace was designed in Fusion 360 using a 3D anatomical knee model and then fabricated through additive manufacturing. It provides a lightweight, personalized fit and securely houses the MPU6050 and PVDF sensors near the patellar region, ensuring accurate real-time monitoring during motion.

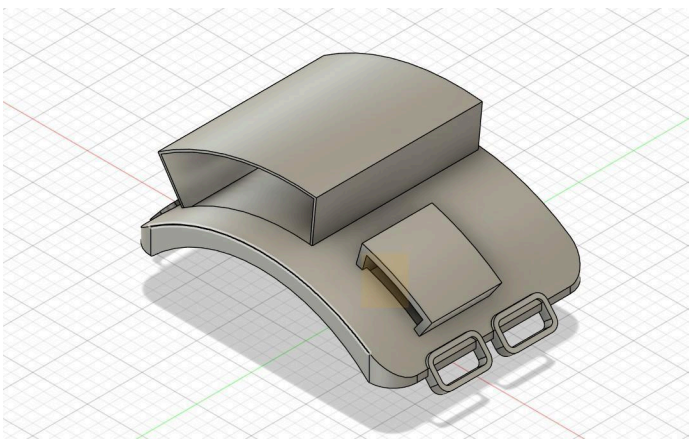


Fig 2.4 3D model of brace

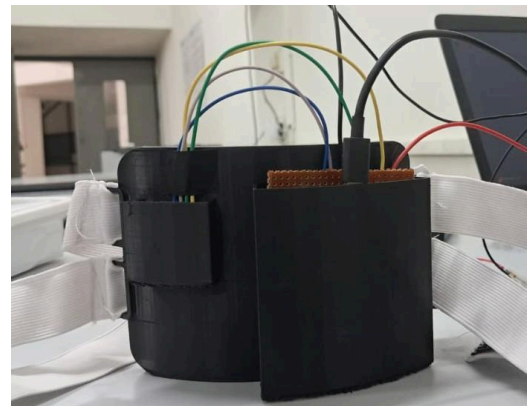


Fig 2.5 Final 3D printed brace

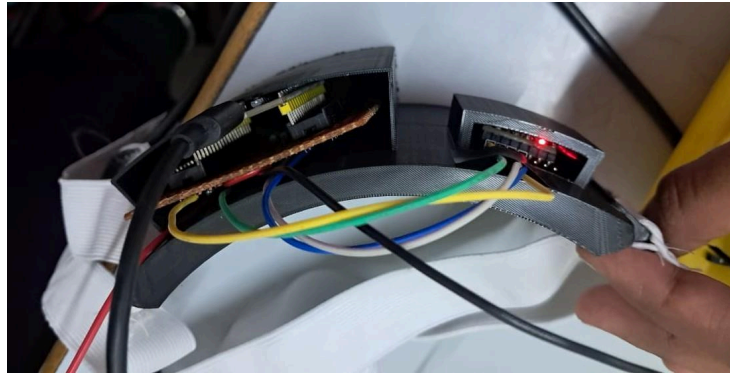


Fig 2.6 Brace housing sensor and microcontroller

PASSIVE COMPONENTS

A $4.7\text{ k}\Omega$ resistor is used as a pull-up resistor to ensure stable and reliable signal communication between the microcontroller and the sensors. It maintains the signal line at a defined HIGH state when no active input is present, preventing floating or undefined values.

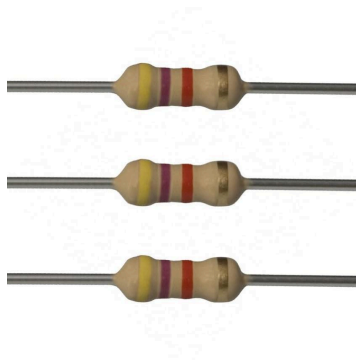


Fig 2.7 4.7K ohm resistor

JUMPER WIRES

It's easy to make connections between electronic components. They are male-to-male, female-to-female, or male-to-female. jumper wires are used to connect components.

2.2 METHODOLOGY

2.2.1 METHODOLOGY OVERVIEW

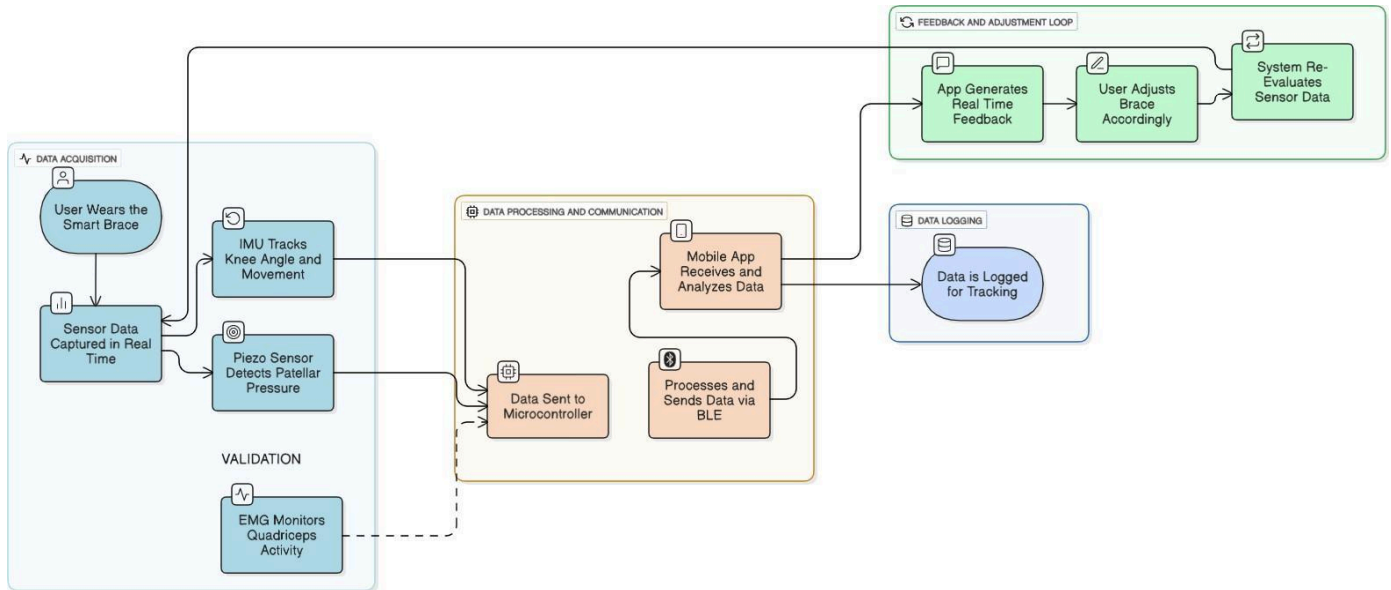


Fig. 2.8 Breakdown of Methodology

The MPU6050 IMU continuously measures the angular movement of the knee, while the PVDF piezoelectric film captures patellar vibration patterns from within the custom 3D-printed brace. These real-time signals are acquired and processed by the ESP32 WROOM microcontroller, which converts the analog and digital inputs into a wireless data stream. The processed data is then transmitted via Wi-Fi to a custom web application.

During initial calibration, the web app sets safe threshold limits for angle and vibration. Once activated, it visualizes the live biomechanical data and actively compares it against these preset values. If any abnormal deviation or excessive patellar vibration is detected, the system immediately generates a warning notification, indicating possible instability or malalignment of the patella.

2.2.2 SIMPLIFIED BLOCK DIAGRAM

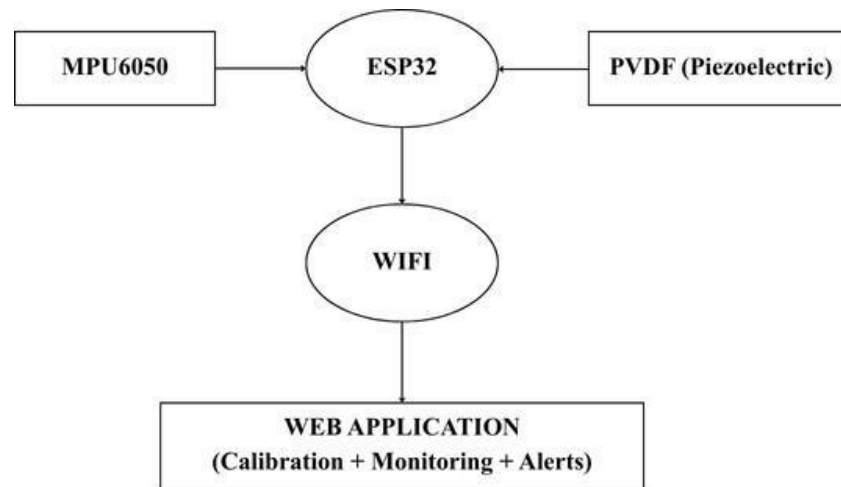


Fig. 2.9 Simplified block diagram of Methodology

2.2.3 SYSTEM WORKFLOW

Signal Acquisition and Data Filtering

The MPU6050 IMU continuously measures knee joint angle, while the PVDF piezoelectric film captures patellar vibration. Both signals are acquired by the ESP32 WROOM microcontroller in real time. The raw PVDF signal is analog in nature and is converted to digital format by the ESP32. Basic filtering is applied to remove noise and ensure stable readings before transmission.

Wireless Transmission

The ESP32 transmits the processed angle and vibration data to the custom web application via Wi-Fi at regular time intervals.

Calibration and Threshold Mapping

During initial setup, the web app performs calibration to record normal knee movement and vibration patterns. From this, safe threshold limits are defined for angle deviation and vibration intensity.

Real-Time Monitoring

The live sensor data is continuously compared with the calibrated threshold values.

Warning and Feedback Mechanism

If the system detects any abnormal increase in vibration or an unsafe angular deviation, an **instant warning alert** is triggered on the web application, indicating potential patellar misalignment or instability.

2.3 CIRCUIT DIAGRAM

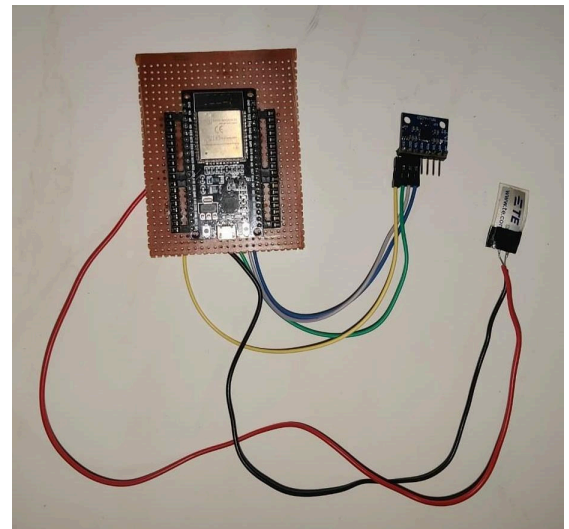
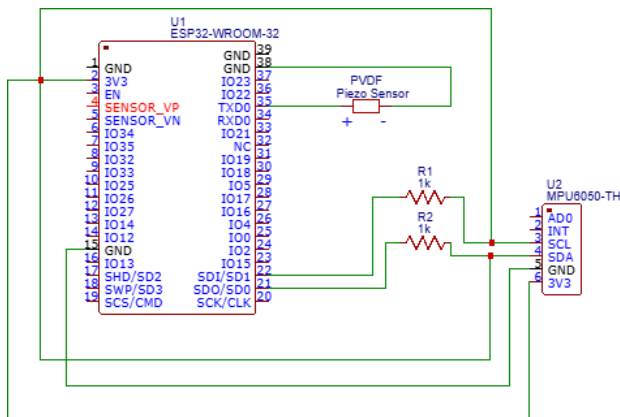


Fig. 2.10 Circuit Design and Circuit Connection of the model

2.3.1 CIRCUIT CONNECTION

Component	ESP32 Pin	Notes
MPU6050 SDA	GPIO21	I2C with 4.7k Ω pull-up
MPU6050 SCL	GPIO22	I2C with 4.7k Ω pull-up
PVDF Sensor	GPIO35	ADC input
MPU6050 VCC	3.3V	Power supply
Ground	GND	Common ground

Table 2.3.1 Circuit connections and respective components

2.4 DATA ACQUISITION



a)With strap



b)Without strap

Fig 2.11 Data acquisition with and without strap

The data acquisition process involved collecting both inertial and piezoelectric sensor data under two primary conditions: with the brace and without the brace. The IMU module was used to measure joint angles during flexion and extension movements, enabling precise assessment of the knee's range of motion in both scenarios. Piezoelectric sensor was employed to capture variations in vibration responses corresponding to the same movements, providing insight into the mechanical interaction between the brace and the limb. All sensor data were interfaced through the ESP32 microcontroller, with real-time visualization performed using the Arduino IDE Serial Plotter. The recorded IMU and piezo data were displayed in the Serial Monitor and saved in '.csv' format for subsequent processing and analysis. Each trial was repeated multiple times under identical conditions to ensure measurement consistency and reliability.

2.4.1 DATA ACQUIRED DURING KNEE FLEXION EXTENSION CYCLES

The table shows synchronized timestamps, knee joint angles, PVDF vibration sensor readings, and activity labels. Data was sampled at ~3.8 Hz using an MPU6050 and a PVDF sensor on an ESP32. Angle was estimated with a complementary filter, while vibration was quantified using a 200-sample RMS window. The dataset highlights clear differences in both motion and vibration patterns between flexion and extension, especially during transitions.

Time stamp	Angle (°)	Vibrations (mV)	Activity
26:36.5	59.9	239.7	Flexion
26:36.8	63.4	523.2	Flexion
26:37.0	65.1	545.3	Flexion
26:37.3	65.5	218.7	Flexion
26:44.6	82.2	333.1	Extension
26:44.8	75.9	527.7	Extension
26:45.1	69.4	427.2	Extension
26:45.3	66.1	346.9	Extension

Table 2.4.1a Excerpt from dataset acquired “With Strap”

Time stamp	Angle (°)	Vibrations (mV)	Activity
57:52.1	61.5	478.5	Flexion
57:52.3	65.7	916.8	Flexion
57:52.8	70.8	979.1	Flexion
57:53.1	72.8	986.9	Flexion
57:53.3	73.5	193.4	Flexion
57:56.4	81.3	517.7	Extension
57:56.6	74.8	714.8	Extension
57:56.9	65.3	1399.8	Extension
57:57.1	62	1406.2	Extension
57:57.3	53.9	1472.3	Extension

Table 2.4.1b Excerpt from dataset acquired “Without Strap”



Fig. 2.12 Knee Angle vs. Sample Index (Without Strap vs. With Strap) during flexion and extension

Knee Angle vs. Time

Without strap: High-frequency oscillations, showing abrupt jumps and less control.

With strap: Smoother waveform and fewer sharp changes, indicating more controlled movement and stabilizing effect of the brace.

The maximum knee angle is also restricted from 94.1° to 87.8° , consistent with brace-imposed ROM limits.

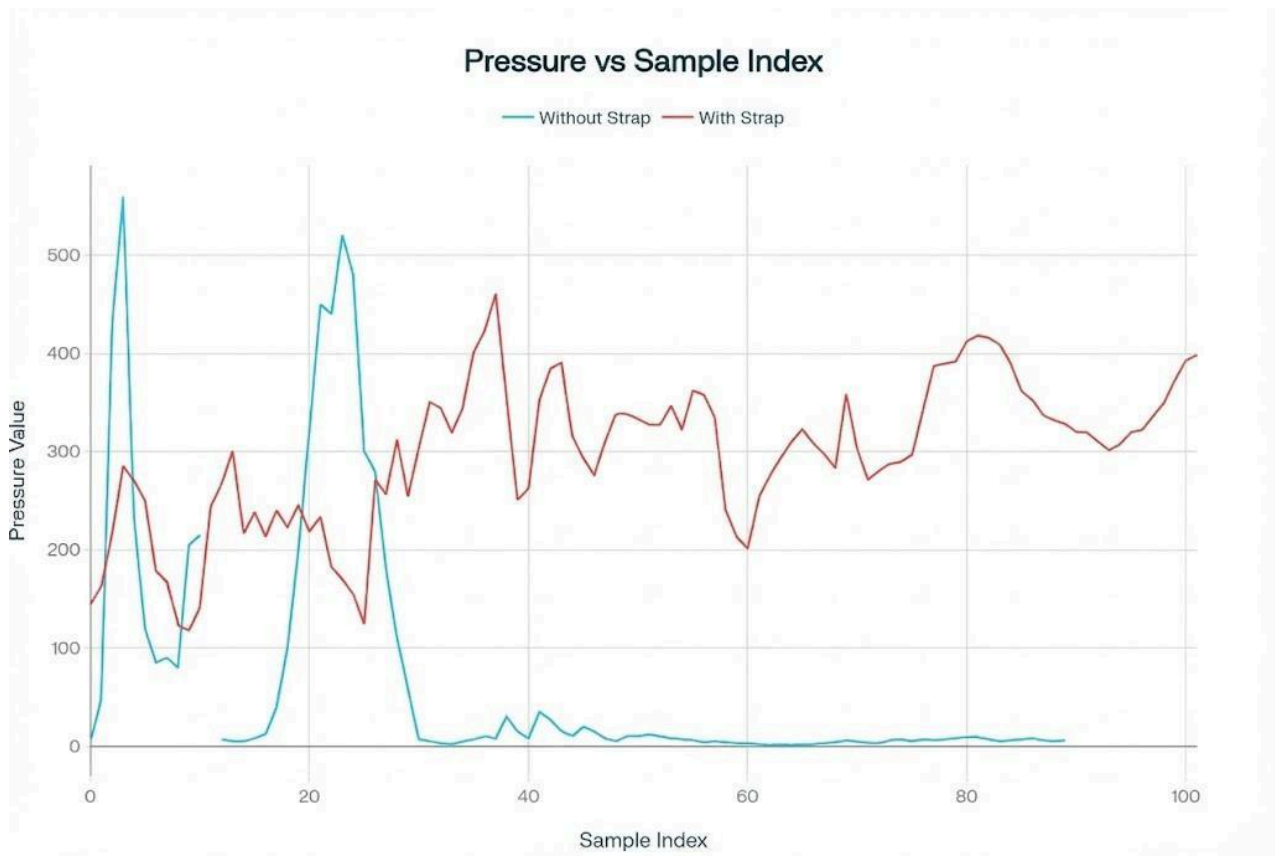


Fig 2.13 Patellar vibrations (mV) vs. Sample Index (Without Strap vs. With Strap) during flexion and extension

INTERPRETATION

Vibration (Piezo RMS) vs. Time

Without strap: Prominent sharp spikes, reflecting large vibration events and mechanical instability.

With strap: Caps these peaks, baseline is higher but sharp spikes are reduced, confirming the brace's damping and stabilization.

Maximum vibration drops substantially (from 1472.3 to 545.3) supporting the claim that vibration spikes are suppressed by the brace, showing effective stabilization.

2.5 DATA PROCESSING

The data acquisition system integrates two complementary sensor modalities to capture comprehensive biomechanical information during knee joint motion. Raw sensor signals undergo systematic preprocessing to enhance signal quality and extract physiologically relevant parameters.

2.5.1 Sensor Data Collection

MPU6050 Inertial Measurement Unit (IMU)

The MPU6050 6-axis IMU provides real-time kinematic data through I²C communication protocol (GPIO21: SDA, GPIO22: SCL). Configured with an accelerometer range of $\pm 8g$ and gyroscope range of $\pm 500^\circ/s$, the sensor captures tri-axial acceleration and angular velocity at a bandwidth of 20 Hz to filter high-frequency mechanical noise while preserving motion dynamics.

PVDF Piezoelectric Vibration Sensor

The polyvinylidene fluoride (PVDF) film sensor, connected to GPIO35 (12-bit ADC input), operates in dynamic strain sensing mode. Mechanical vibrations generated during joint movement induce charge separation in the piezoelectric material, producing voltage proportional to vibration amplitude. The sensor captures transient impact forces and joint stability indicators with high temporal resolution.

Complementary Filter Implementation

To obtain accurate knee angle estimation, a complementary filter fuses accelerometer and gyroscope data, leveraging the strengths of both sensors while compensating for their individual limitations.

Algorithm:

$$\theta_{\text{filtered}} = \alpha \times (\theta_{\text{previous}} + \omega_{\text{gyro}} \times \Delta t) + (1 - \alpha) \times \theta_{\text{accel}}$$

(Eqn. 2.1)

where:

θ_{filtered} = current filtered angle

θ_{previous} = previous angle estimate

ω_{gyro} = angular velocity from gyroscope ($^{\circ}/\text{s}$)

Δt = sampling interval (0.265 seconds)

θ_{accel} = angle calculated from accelerometer: $\tan^{-1} (a_y / a_z) \times (180/\pi)$

α = filter coefficient (0.98)

The high-pass filtered gyroscope component (98% weight) provides smooth, drift-free short-term tracking, while the low-pass filtered accelerometer component (2% weight) corrects long-term drift, yielding stable angle measurements resistant to both integration drift and acceleration artifacts.

RMS Vibration Calculation

Joint vibration amplitude is quantified using Root Mean Square (RMS) analysis, which provides a scalar measure of vibration intensity independent of signal polarity.

Algorithm

$$V_{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2}$$

(Eqn 2.2)

where:

- N = 200 samples (collected over ~200 ms)
- X_i = individual ADC reading (0–4095)
- $\bar{X}$ = mean ADC value across the sampling window

Voltage calculation

The piezo sensor produces voltage that is read by the ESP32's 12-bit ADC (range 0–4095). The system collects 200 samples spaced 1 millisecond apart. The Root Mean Square (RMS) of these samples is calculated to quantify vibration strength.

The RMS ADC value is converted to millivolts using:

$$\text{Voltage (in mV)} = (\text{RMS ADC} / 4095) \times 3300$$

(Eqn.2.3)

A noise threshold of 10 millivolts is applied to remove low-level electronic noise.

2.5.2 Sensor Data Processing steps

1. Sample Collection: Acquire 200 consecutive ADC readings at 1 ms intervals
2. Mean Calculation: Compute DC offset ($\bar{x}$)
3. Centering: Subtract mean from each sample to remove baseline
4. Squared Sum: Calculate $\sum (x_i - \bar{x})^2$
5. RMS Extraction: Take square root of mean squared value
6. Voltage calculation: Calculate millivolt value from RMS value

This approach isolates AC vibration components from DC sensor bias, providing a robust metric for comparing joint stability across experimental conditions.

2.5.3 Data Analysis and Visualization

The piezoelectric sensor data was recorded to measure variations in vibration across the knee joint during flexion and extension movements, both with and without the brace. Simultaneously, IMU data was collected to monitor angular changes and movement patterns associated with patellar tracking. The acquired signals were preprocessed to remove unwanted noise using:

1. Complementary filtering (98% gyro + 2% accel) for angle estimation
2. 200-sample RMS calculation for vibration quantification
3. Noise baseline removal (threshold: 20 units)

ensuring accurate interpretation of mechanical responses. The processed piezo data was plotted to visualize vibration fluctuations over time, while the IMU data was analyzed to determine angular displacement and joint motion characteristics.

2.5.4 INSIGHTS

Metric	Without Strap	With Strap
Mean knee angle	46.7	63.1
Std deviation knee angle	24.2	17
Max knee angle	94.1	87.8
Min knee angle	8.2	24
Mean vibration (mV)	251.73	298.36
Std deviation (mV)	361.02	100.02
Max vibration (mV)	1472.3	545.3
Min vibration (mV)	0	117.3
Angle-vibration correlation	-0.016	0.315
RMS	440.12	214.68

Table 2.5.2 INSIGHTS

2.5.5 Identification of Maltracking

A two-phase calibration is used for individual baseline setting:

- Phase 1 (0–10s): Stationary period to average initial knee angle, establishing a zero reference.
- Phase 2 (10–30s): User performs controlled knee movements. The system records dynamic range and typical vibration levels, setting safe thresholds for mechanical deviation.

Thresholds are calculated after calibration:

- **Angle safe limit:** The highest angle observed during movement is noted, and a safety margin of 10 degrees is added. This value defines the point above which the knee angle is considered excessive.
- **Vibration safe limit:** All vibration readings from the movement phase are used to compute their average value (mean) and their variation (standard deviation).

The safe vibration limit is defined as:

$$\text{vibration safe limit} = \text{mean vibration} + 2 \times \text{standard deviation}$$

[10] (Eqn 2.4)

Values above this are treated as unusually high vibration, suggesting increased patellofemoral load.

Monitoring logic:

During real-time use, the system performs two checks:

1. **Relative angle = measured angle – angle offset.** If the relative angle exceeds the angle safe limit, an angle alert is triggered.
2. If the vibration level exceeds the vibration safe limit, a vibration alert is triggered.

2.5.6 Alert Mechanism

The system was designed to continuously monitor the real-time piezo and IMU readings through the ESP32 microcontroller. When the data exceeded a defined threshold representing abnormal vibrational or angular deviation, the system triggered an alert signal to notify the user of potential patellar misalignment. This immediate feedback mechanism supports safer movement and helps maintain proper joint alignment during rehabilitation or daily activities.

CHAPTER 3

3.1 RESULT AND DIISCUSSION

3.1.1 OUTPUT

The final output we obtained in our project is a web app that acquires sensor information via the ESP32 and alerts individuals in case abnormal patellar vibrations are detected.

i) ESP32 connects to app

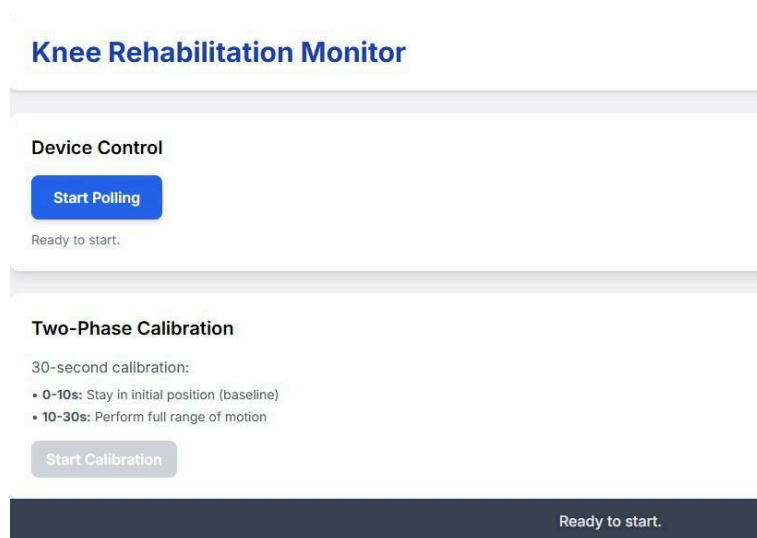


Fig 3.1 ESP32 connects to app

ii) Angle Calibration for user

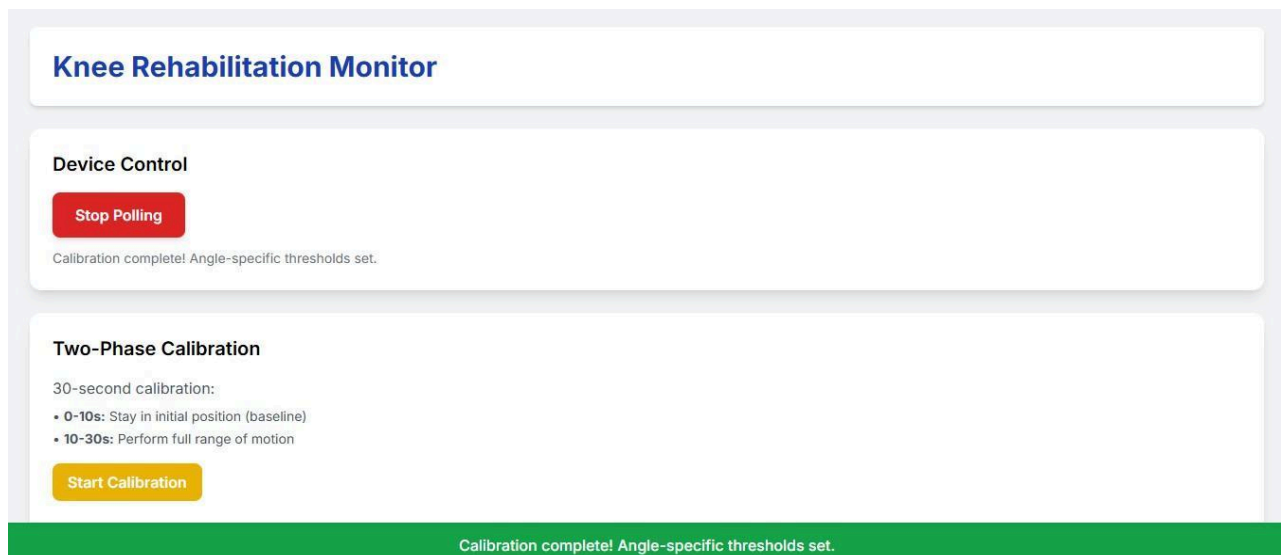


Fig 3.2 Angle Calibration for user

iii) Knee angle and vibration displayed

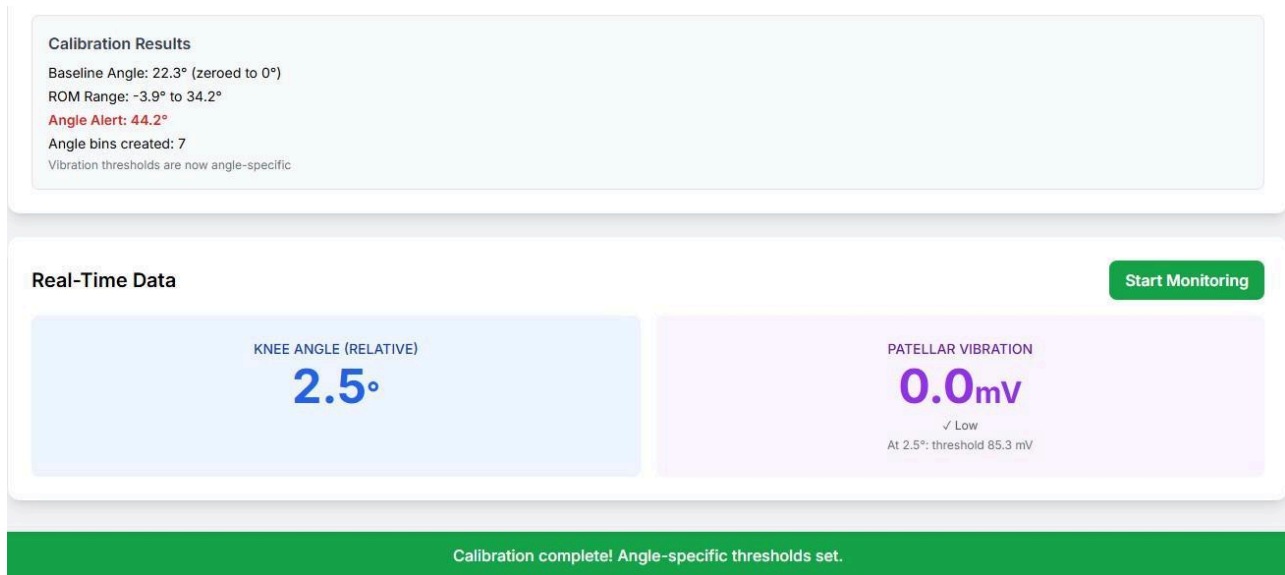


Fig 3.3 Knee angle and vibration displayed

iv) Abnormal vibration detected and alerted

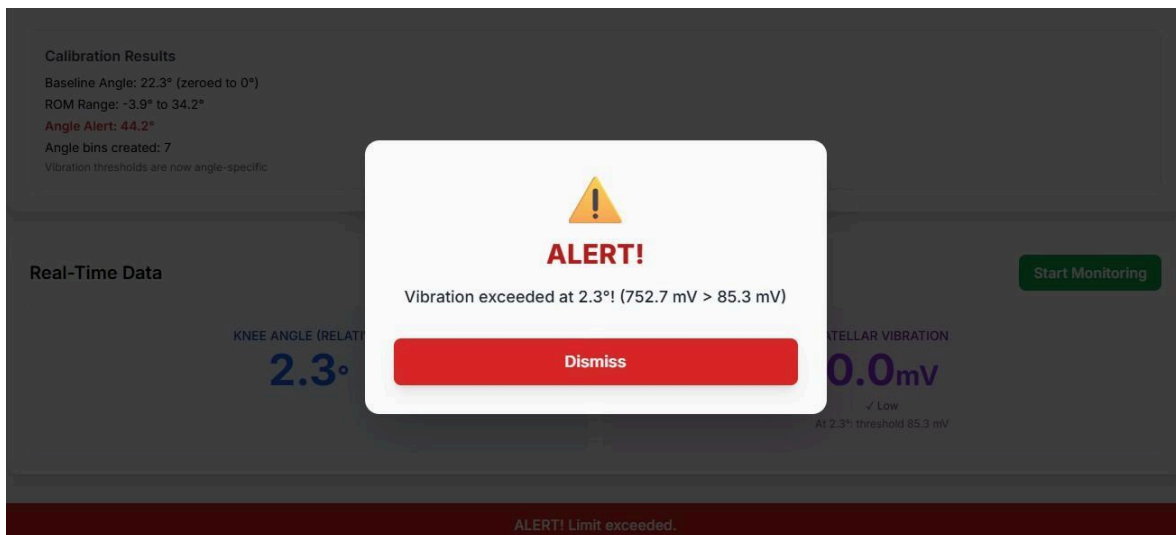


Fig 3.4 Abnormal vibration detected and alerted

3.1.2 DISCUSSIONS

The analysis of knee angle and piezo vibration data reveals a clear stabilizing effect of the brace. The mean knee angle was higher with the strap (63.1°) compared to without strap (46.7°), likely because the “with strap” measurements began after partial extension; nevertheless, the values lie within the functional range of motion. The standard deviation of knee angle decreased from 24.2° to 17.0° with the brace, indicating smoother, more controlled flexion-extension. Maximum knee angle was reduced from 94.1° to 87.8° , and minimum angle increased from 8.2° to 24.0° , reflecting restricted extreme motion and a new neutral alignment imposed by the brace.

Voltage analysis demonstrates that mean vibration increased with the brace (251.73 vs 298.36) due to the constant baseline compression applied by the strap. Variability in vibrations decreased substantially (standard deviation 361.02 vs 100.02, while maximum vibration dropped from 1472.3 to 545.3 and minimum vibration increased from near zero to 117.3, indicating more uniform load distribution and effective vibration damping. The RMS vibration reduced from 440.12 to 214.68, and a slight positive correlation (0.31) between knee angle and vibration suggests that increased flexion leads to controlled rises in patellar load. These results confirm that the “with strap” condition significantly reduces micro-vibrations and smooths angular motion.

Mechanically, the brace acts as a viscoelastic damping layer, absorbing energy from rapid knee flexion-induced vibrations before it reaches the piezo sensor, reducing RMS amplitude. The smoother IMU signals indicate improved motion stability, with better control of joint articulation and reduced lateral displacement. Clinically, this confirms that the patient-specific 3D-printed brace provides stabilization without over-restricting movement, supporting safer, controlled rehabilitation for users with patellar maltracking or misalignment.

3.2 INFERENCE

1. The comparative analysis of piezo and IMU signals clearly demonstrates the mechanical influence of the knee brace on joint stability and load distribution. Under unbraced conditions, the piezo sensor exhibited irregular high-magnitude vibration spikes, while the IMU showed abrupt and uncoordinated angular transitions, signifying poor control and unrestrained knee motion. These patterns indicate increased vibration transmission and uneven stress across the patellar surface during flexion and extension.
2. When the brace was applied, the piezo waveform became smoother and maintained a higher baseline vibrations, with significantly reduced amplitude peaks. This reflects improved damping performance, where the brace material absorbs a portion of the mechanical energy generated during motion. Similarly, the IMU output displayed uniform angular progression, indicating stabilized and symmetrical knee articulation under controlled movement conditions.
3. Quantitatively, the RMS and standard deviation values dropped notably in the braced state, confirming reduced vibration and enhanced consistency in load transfer. The correlation between knee angle and piezo vibration further supports the inference that higher flexion corresponds to greater but controlled vibration, demonstrating proper brace functionality. Together, these findings validate the brace's capability to maintain patellar alignment and minimize maltracking through effective mechanical stabilization.

CHAPTER 4

CONCLUSION

4.1 Future Work

Future development of the patient-specific, sensor-integrated rehabilitation brace could focus on enhancing real-time adaptability and predictive feedback through advanced machine learning algorithms. Efforts may be directed toward improving the precision of vibration, and motion detection by refining sensor calibration and data fusion techniques. Expanding the system to support long-term monitoring, automated adjustment of brace tension, and personalized rehabilitation protocols could further optimize patient outcomes. Additionally, integrating cloud-based analytics, multilingual mobile app interfaces, and user-defined exercise programs may increase accessibility and flexibility, enabling more customized and globally applicable rehabilitation solutions.

4.2 Novelty

The use of a patient-specific, 3D-printed brace geometry allows for a tailored fit that conforms closely to individual anatomical features, enhancing both comfort and therapeutic support. Integration of PVDF film sensors and inertial measurement units (IMUs) via an ESP32 WROOM microcontroller enables continuous monitoring vibration, and movement, providing detailed biomechanical data. Wireless transmission of this data to a mobile application facilitates ongoing remote monitoring and feedback, supporting adaptive rehabilitation strategies that can be adjusted according to the patient's progress.

4.3 Conclusion

The experimental evaluation confirms that the 3D-printed, patient-specific knee brace significantly improves knee joint stability by minimizing abrupt angular shifts and vibration-induced pressure fluctuations. The consistent reduction in RMS and peak vibration values under the “with strap” condition indicates that the brace successfully distributes mechanical load, thereby reducing localized stress and the likelihood of patellar misalignment.

The smoother IMU angular patterns under braced motion validate improved control during flexion and extension, proving that the brace not only provides structural support but also enhances biomechanical balance. The continuous contact vibration observed through the piezo sensor suggests that the brace ensures stable alignment without complete restriction of movement, maintaining functional mobility while supporting corrective positioning.

Overall, the integration of piezoelectric and IMU sensing with the ESP32 platform enabled an effective real-time assessment of knee joint biomechanics. The results confirm that the developed brace design assists in damping vibrational energy, controlling motion irregularities, and improving patellar tracking. Hence, this system demonstrates potential for clinical and rehabilitative use as a wearable diagnostic tool for monitoring and managing patellar maltracking.

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